

Figure 1: Control Power hardware via RS232 cable

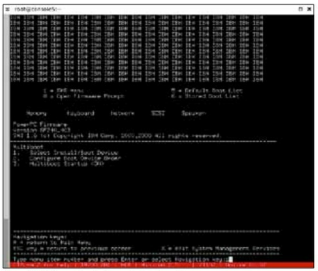


Figure 2: SMS greeting before entering OpenFirmware menu

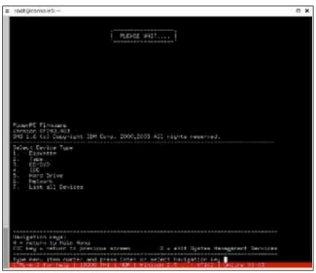


Figure 3: Choose installation source

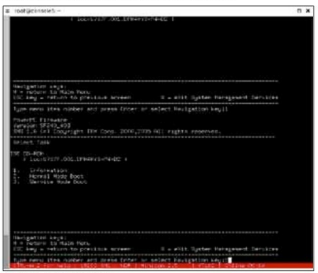


Figure 4: Choose boot mode

Remote access

IBM produces equipment that has OpenFirmware (OF) inside. This is analogous to the traditional PC BIOS used on x86 systems. With OF's help, you can assign which source should be used to start an OS, perform a number of diagnostic procedures, etc. OF itself uses a ServiceProcessor—a special unit embedded into the big tower. When you plug in the power cord, first the ServiceProcessor starts. That initially sets up a remote access option, and then awaits the user's response. The user has to press the hardware Start button on the front panel. So, when plugged in, IntelliStation is never completely turned off—even with the main CPUs off, the ServiceProcessor doesn't sleep. The scheme is reminiscent of a similar product—iLO [1], developed by Compaq/HP. ILO also lets one start and control a server remotely. It's worth noting that IBM also has additional functions run by HMC (Hardware Management Console), among which are planning and allocation of physical resources to logical partitions (LPAR [3]/DLPAR [4]), splitting main CPU performance onto logical CPUs (Capacity On Demand) and related actions that provide virtualisation management on POWER-based processors. Unfortunately, I can only mention these capabilities, and not go any deeper into the subject, because that would be beyond the scope of this review.

So, in order to establish an access link to the ServiceProcessor and later install Linux, you will need an RS232(F)<->RS232(F) cable and a dedicated workstation, from which you can directly control your IntelliStation 285. As a console, any PC with an installed command-line tool like *minicom* or *surecom* (a GUI alternative) will suit. Set the communication speed to 19200 bit/s, the standard default rate for for IBM hardware, plug in the power cord and wait for the login prompt after running *minicom -D /dev/ttyS0 -s 19200 -o*. See Figure 1.

To start the whole system and gain access to OF, from where you can select an OS source, choose: *1. Power/Restart*